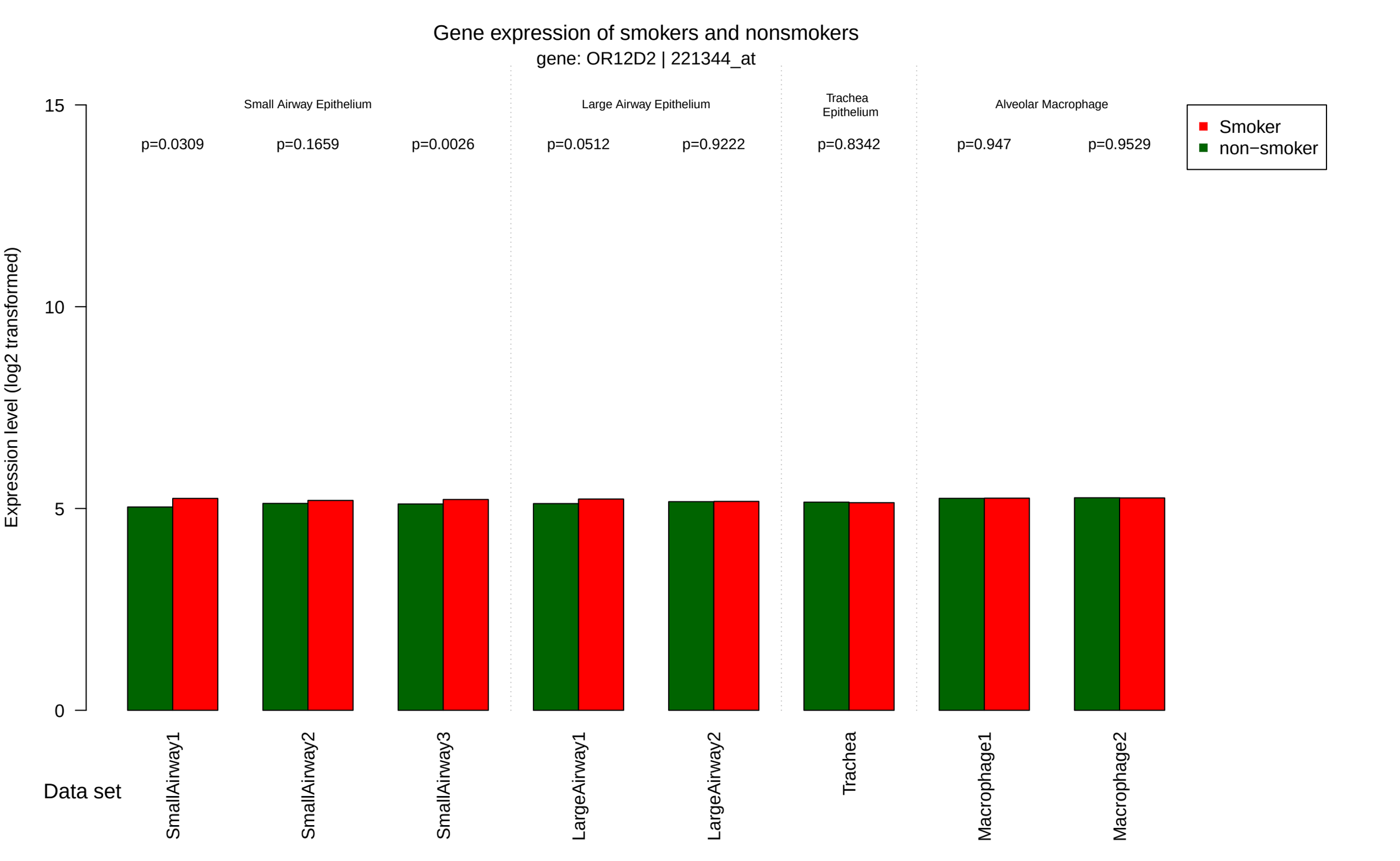
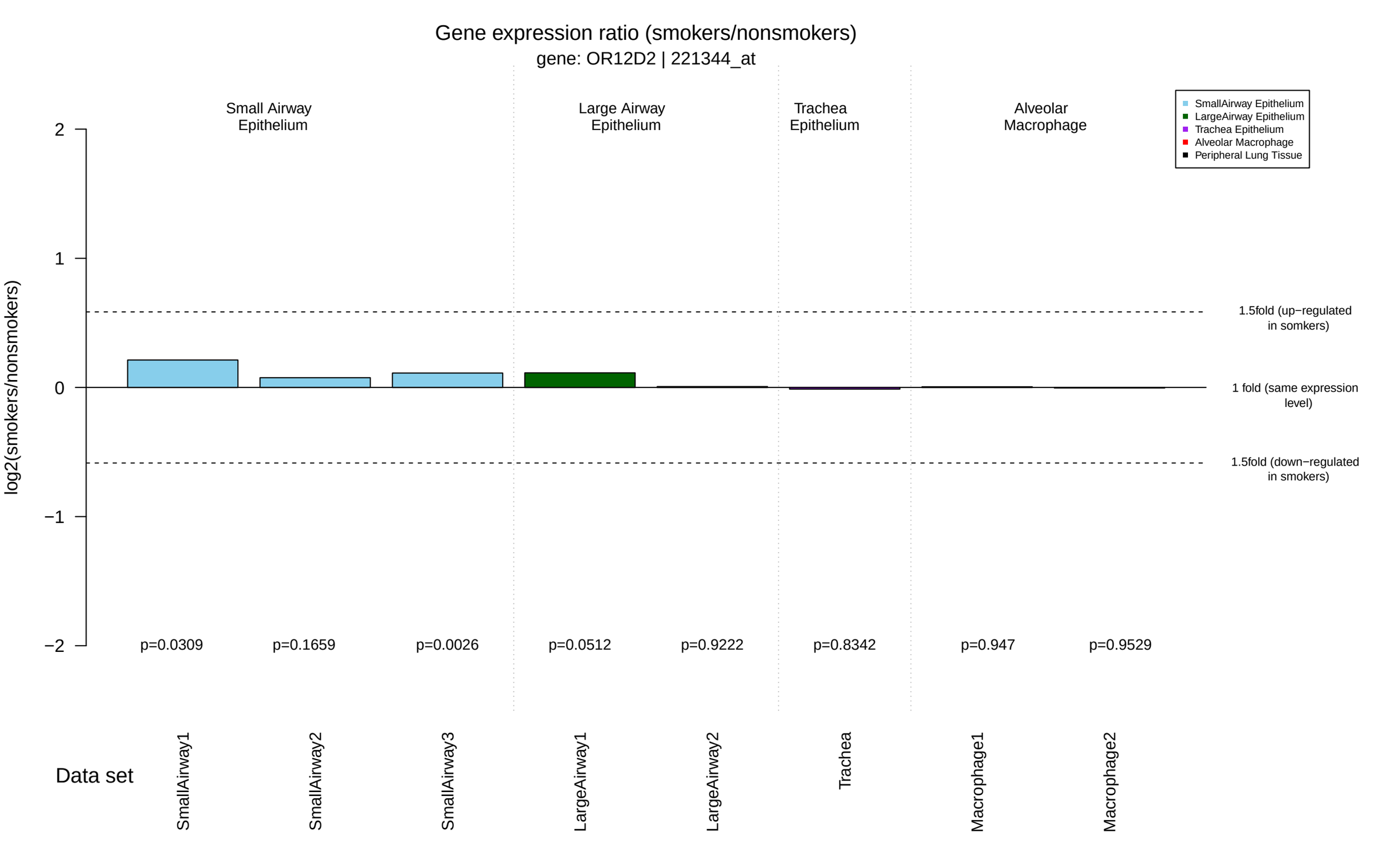
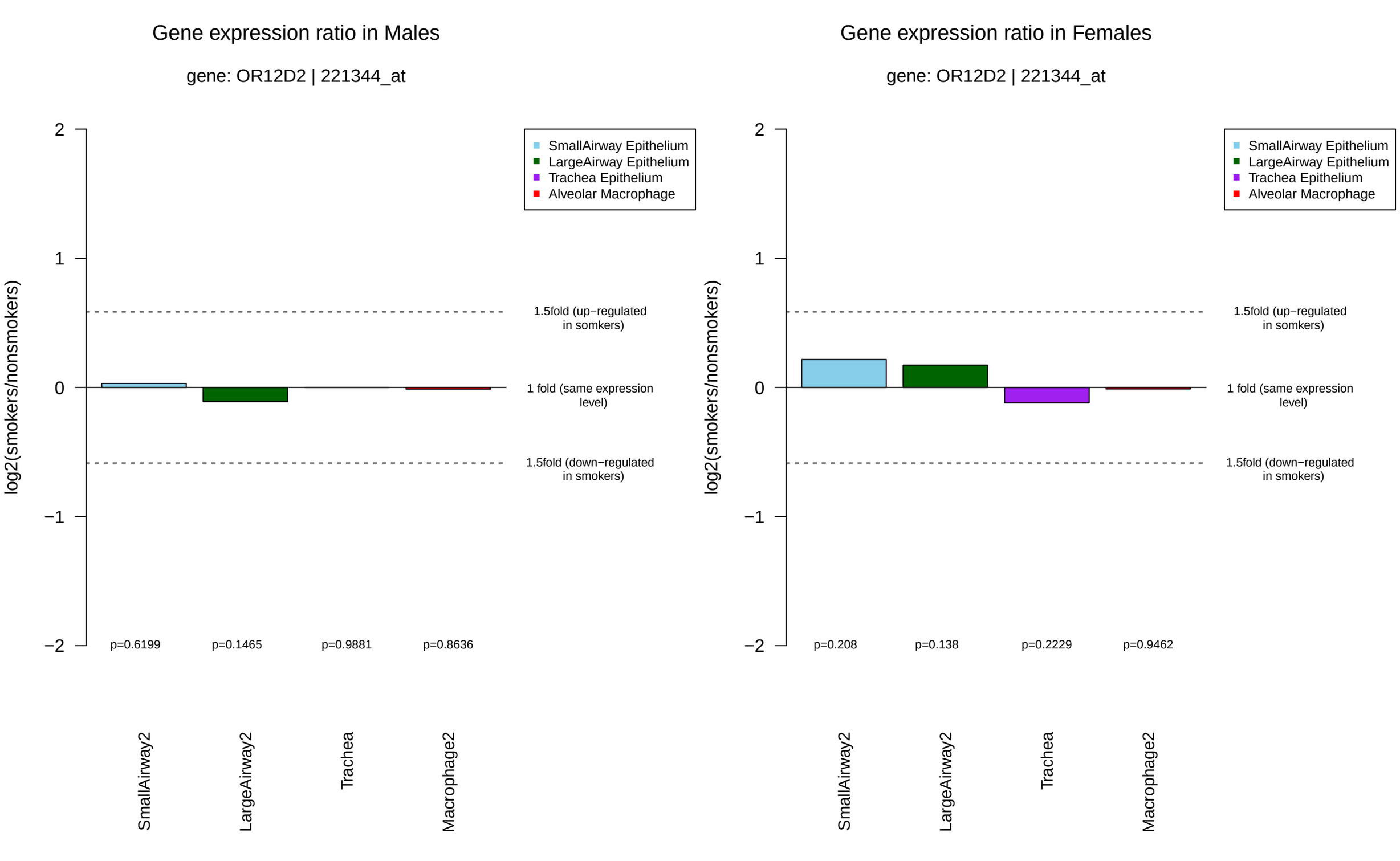
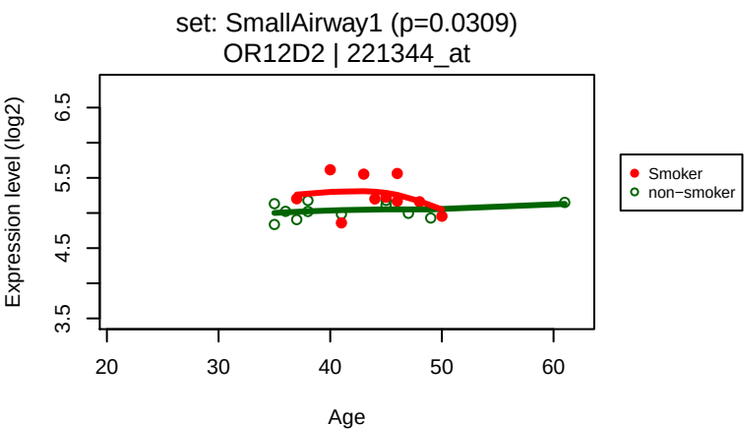
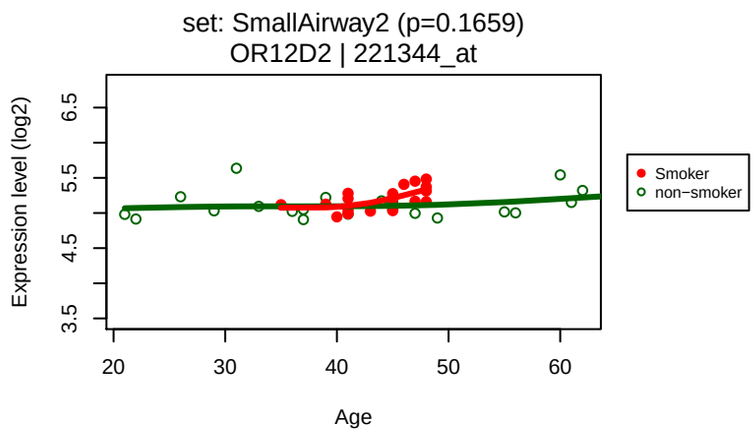
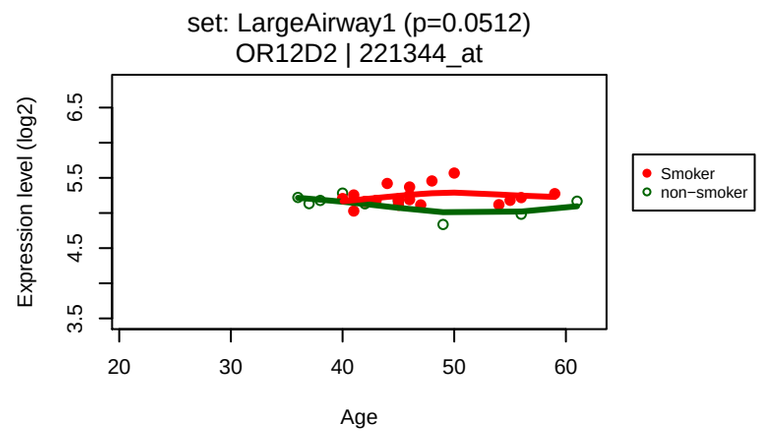

Gene expression of smokers and nonsmokers
gene: OR12D2 | 221344_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0309
p=0.1659
p=0.0026
p=0.0512
p=0.9222
p=0.8342
p=0.947
p=0.9529
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

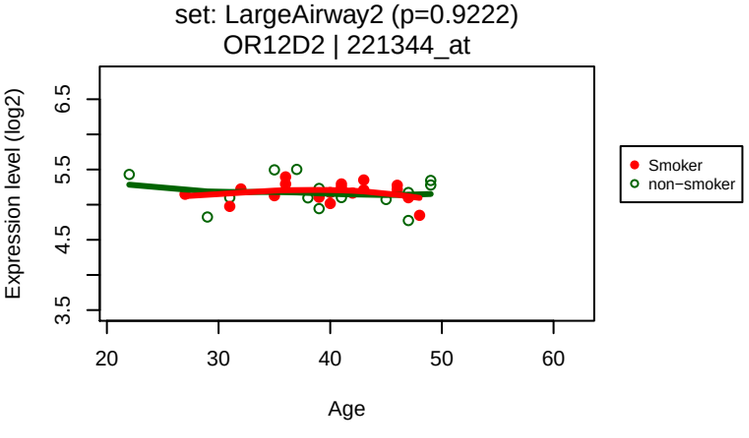

Gene expression ratio (smokers/nonsmokers)
gene: OR12D2 | 221344_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0309
p=0.1659
p=0.0026
p=0.0512
p=0.9222
p=0.8342
p=0.947
p=0.9529
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

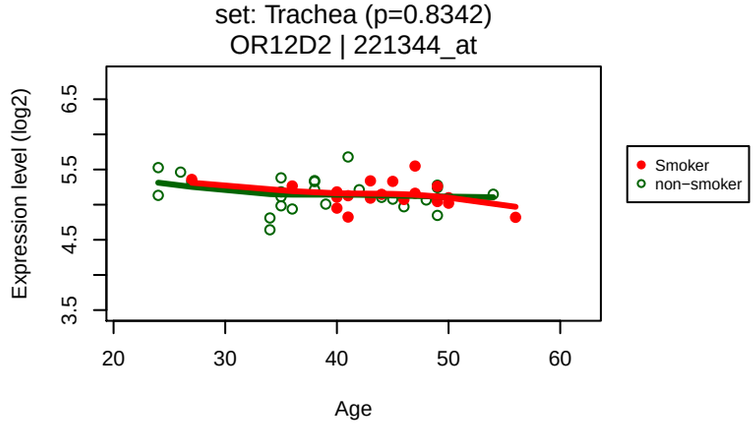

Gene expression ratio in Males
gene: OR12D2 | 221344_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6199
p=0.1465
p=0.9881
p=0.8636
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: OR12D2 | 221344_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.208
p=0.138
p=0.2229
p=0.9462
SmallAirway2
LargeAirway2
Trachea
Macrophage2

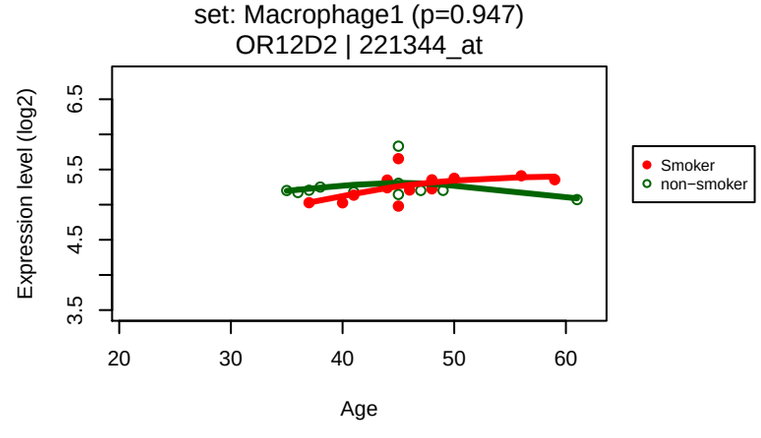

set: SmallAirway1 (p=0.0309)
OR12D2 | 221344_at
Expression level (log2)
Age
Smoker
non−smoker

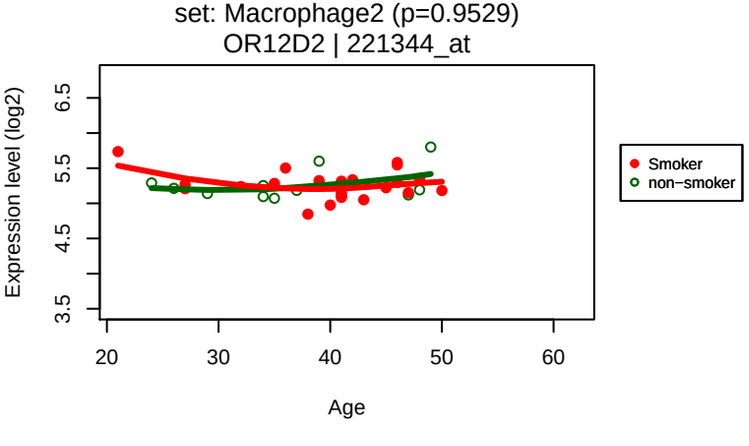

set: SmallAirway2 (p=0.1659)
OR12D2 | 221344_at
Expression level (log2)
Age
Smoker
non−smoker

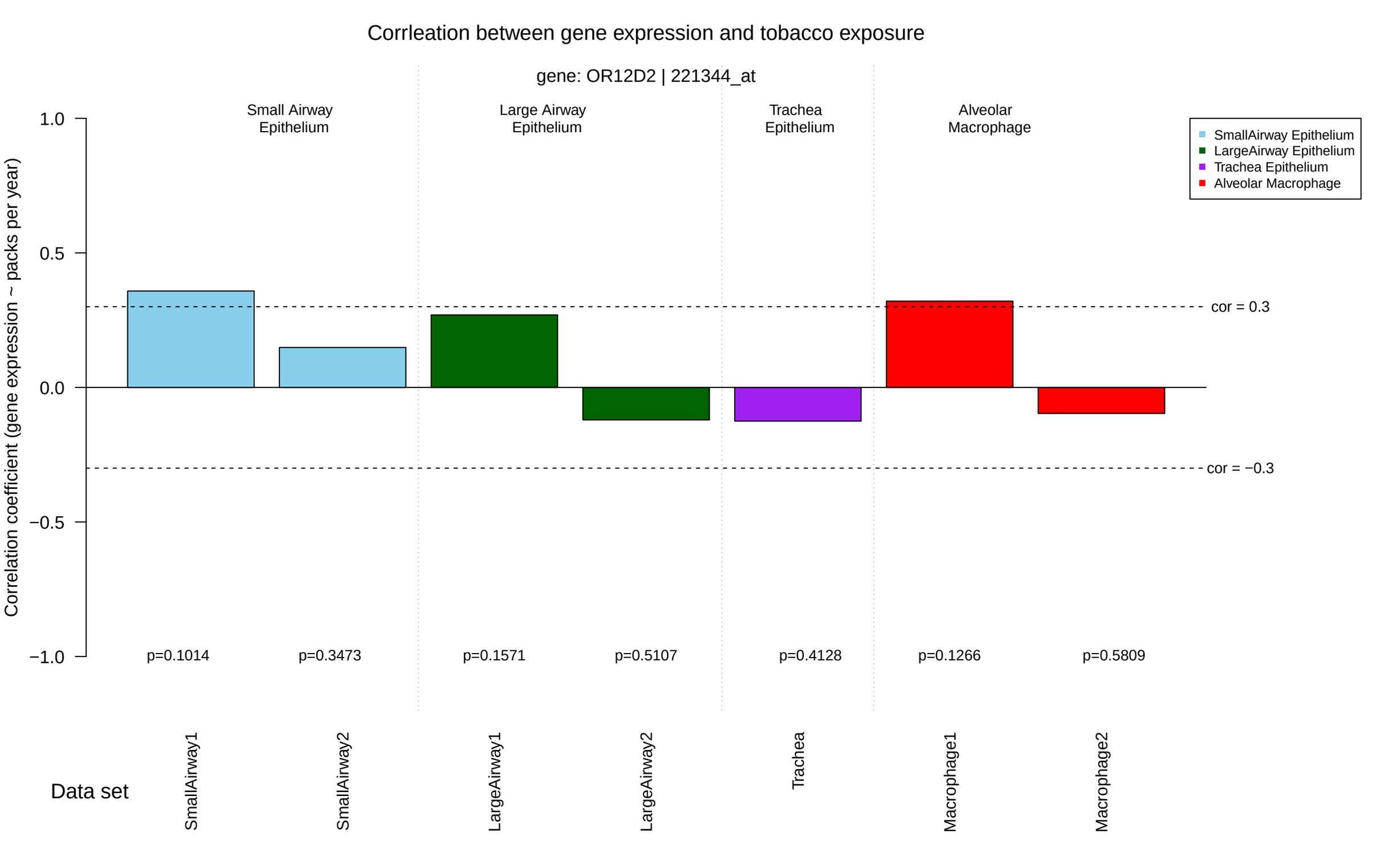

set: LargeAirway1 (p=0.0512)
OR12D2 | 221344_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9222)
OR12D2 | 221344_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.8342)
OR12D2 | 221344_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.947)
OR12D2 | 221344_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.9529)
OR12D2 | 221344_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: OR12D2 | 221344_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1014
p=0.3473
p=0.1571
p=0.5107
p=0.4128
p=0.1266
p=0.5809
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2